

1a) $f(x)$ is continuous on $[-\frac{1}{2}, \frac{3}{4}]$ and therefore the Closed Interval Method can be used.

$$f(x) = \frac{2x^2}{(x^2-1)}, \quad f'(x) = \frac{(x^2-1)4x - 2x^2(2x)}{(x^2-1)^2} = \frac{-4x}{(x^2-1)^2}$$

Since $f'(x)$ exists for all $x \in [-\frac{1}{2}, \frac{3}{4}]$, the only critical numbers of f occur when $f'(x) = 0$, that is $x = 0$.

$$f(0) = 0$$

at the end points

$$f\left(-\frac{1}{2}\right) = \frac{2\left(-\frac{1}{2}\right)^2}{\left(-\frac{1}{2}\right)^2 - 1} = \frac{\frac{1}{2}}{\frac{1}{4} - 1} = \frac{1}{2} \cdot \frac{4}{3} = \frac{2}{3}$$

$$f\left(\frac{3}{4}\right) = \frac{2\left(\frac{3}{4}\right)^2}{\left(\frac{3}{4}\right)^2 - 1} = \frac{\frac{9}{8}}{\frac{9}{16} - 1} = \frac{\frac{9}{8}}{-\frac{7}{16}} = \frac{9}{8} \cdot \frac{16}{7} = \frac{18}{7}$$

$$\text{abs max} - f(0) = 0$$

$$\text{abs min} - f\left(\frac{3}{4}\right) = -\frac{18}{7}$$

b) From (1a) $f'(x) = \frac{-4x}{(x^2-1)^2}$, $f'(x)$ does not exist for $x = -1, 1$
 $f'(x) = 0$ for $x = 0$

f is not continuous on $[-\frac{1}{2}, 2]$ therefore the closed interval method can not be used.

$$\lim_{x \rightarrow 1^+} \frac{2x^2}{(x^2-1)} = +\infty, \quad \lim_{x \rightarrow 1^-} \frac{2x^2}{(x^2-1)} = -\infty, \quad f(0) = 0$$

Hence there is no absolute min or max

c) Since $f(x)$ is continuous over $[-2, 2]$ we can use the closed interval Method

$$f'(x) = 5(x+1)^4 - 5$$

Since $f'(x)$ exists for all $x \in [-2, 2]$ all critical numbers occur where $f'(x) = 0$

$$\begin{aligned} 5(x+1)^4 - 5 &= 0 \\ (x+1)^4 - 1 &= [(x+1)^2 - 1][(x+1)^2 + 1] = 0 \\ &= [(x+1)-1][(x+1)+1][(x+1)^2 + 1] = 0 \\ &= x(x+2)[(x+1)^2 + 1] = 0 \end{aligned}$$

$$f'(x) = 0 \text{ when } x = 0 \text{ or } x = -2.$$

at the critical points $f(0) = -3$

$$f(-2) = 7$$

At the endpoints of I $f(2) = 23$
 $f(5) = 15$

$$f(2) = 1^5 + 5 \cdot 2 - 2$$
$$= 7$$

The abs. max is $f'(2) = 231$

The abs min is $f(0) = -1$.

2a) $f'(x) = 2x - 4$ $f'(x) > 0$ for $x > 2$
 $f'(x) < 0$ for $x < 2$

$\therefore f$ is increasing for $x > 2$

f is decreasing for $x < 2$

$$b) f'(x) = 1 + \frac{1}{x^2} = \frac{x^2 + 1}{x^2} = \frac{(x+1)(x-1)}{x^2}$$

$$f'(x) > 0 \text{ for } x \in (-\infty, -1) \cup (1, \infty) \text{ and } f'(x) < 0 \text{ for } x \in (-1, 1)$$

$f(x)$ is increasing for $x \in (-\infty, -1) \cup (1, \infty)$ and

$f(x)$ is decreasing for $x \in (-1, 1)$, $x \neq 0$

c) $f'(x) = 3x^2$

c) $f'(x) = 3x^2$
 $f'(x) > 0$ for all x therefore $f(x)$ is increasing
 for all x .

3 a) $f'(x) = 2x - 4$ and $f'(x) = 0$ when $x = 2$

using 2a $f(2) = -10$ is a local min value

There is no local max

b) From 2b $f'(x) = 0$ for $x = 1, -1$ - the min/max points
 $\begin{matrix} \text{min} & \text{dec} & \text{inc} & \text{max} \end{matrix}$ $f(1) = 0$

$f(x)$

inc -1 dec 1 inc

Local min $f(1) = 2$
Local max $f(-1) = -2$

Local max $f(-1) = -2$

c) $f'(x) = 3x^2$, $f'(0) = 0$ but $f'(x) > 0$ for $x > 0$ and $x < 0$
Therefore there is no local min or max point

3 2nd derivative test (3a-3b)

3a $f'(x)=0$ for $x=2$ $f''(x)=2 > 0$
Therefore $f(2)$ is a local min value

b) $f'(x) = 1 - \frac{1}{x^2}$, $f'(x)=0$ for $x=-1, 1$ (possible min/max points)
 $f''(x) = 2x^{-3}$, $f''(-1) = \frac{2}{-1} = -2 < 0$
 $f''(1) = \frac{2}{1} = 2 > 0$

local maximum = $f(-1) = -2$

local minimum = $f(1) = 2$

c) $f'(x)=0$ for $x=0$, $f''(x)=6x$ and $f''(0)=0$.
this test can therefore not be used in this case.

4) critical points occur where $f'(x)=0$ or $f'(x)$ does not exist.

a) $f'(x) = 16x - \frac{2}{2x}$

$16x - \frac{1}{x} = 0$

$x^2 = \frac{1}{16}$

$x = \pm \frac{1}{4}$

$f'(x)$ does not exist
when $x=0$

The only CP is

$x = \frac{1}{4}$ because

$x = -\frac{1}{4}$ and $x=0$
is not in D_f

b) $f'(x) = \frac{\sqrt{x+1} \cdot 2x - x^2 \cdot \frac{1}{2\sqrt{x+1}}}{(x+1)}$
 $= \frac{x(3x+4)}{2(x+1)^{3/2}}$

$f'(x)=0$ for $x=0$ or $x=-\frac{4}{3}$

$x = -\frac{4}{3} \notin D_f$

$x=0$ is the only critical number / critical value

(5)

